

Supplementary Information for:
*Revisiting Hall–Petch strengthening in FCC multi-principal
element alloys: what grain size, composition, and processing
can and cannot explain*

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This Supplementary Information follows the same fixed model-family hierarchy as Fig. 1 of the main paper: Family 1 (Classical Hall-Petch), Family 2 (Physics-derived descriptors), Family 3 (Composition/processing), Family 4 (Non-linear estimators), and Family 5 (Symbolic regression). The names and order remain unchanged in both documents. S1–S4 are cumulative input sets within Family 4, not separate model families. E1–E3 are literature evidence classes and are also separate from the model hierarchy.

Primary internal performance is reported using shuffled 5-fold cross-validation, leave-one-out (LOO), and leave-one-batch-out (LOBO) validation. LOO operates at the record level. When a composition is repeated, another record with the same composition can remain in the training set. We therefore do not describe LOO as strict unseen-composition validation. LOBO tests transfer to a new Bayesian-optimization iteration. External literature testing and singularity audits of closed-form equations are reported separately from internal predictive validation.

S1 Dataset audit and exploratory analysis

S1.1 Campaign provenance, synthesis, and characterization

The broader alloy-discovery effort began with Campaign A, which established the closed-loop, multi-objective workflow [1]. Campaign A is a methodological prequel and does not contribute records to the present property-modeling dataset. This dataset combines the subsequent B and C Bayesian-optimization campaigns carried out at Texas A&M University through the BIRDSHOT initiative within the ARL HTMDEC program [2, 3]. The B campaign contributes 35 records from batches BBA, BBB, and BBC, including 34 with YS. It explored the Al–Co–Cr–Cu–Fe–Mn–Ni–V system under an FCC phase-stability constraint while holding processing fixed. Section 2.2 of Ref. [3] documents the B-campaign synthesis and specimen-preparation workflow.

The C campaign contributes 59 records from CBA, CBB, and CBC. It used insights from the B campaign to focus on V–Cr–Mn–Fe–Co–Ni alloys and jointly varied composition, cold work, recrystallization temperature, and hold time. During design-space curation, a modified JMAK-type static-recrystallization and grain-growth prior provided grain-size estimates for feasible candidates. The complete C-campaign and grain-growth-model details will be reported separately. The present regression analysis uses measured EBSD grain sizes rather than the design-stage estimates.

Both campaigns followed the same high-level route of vacuum-arc melting from high-purity elements, homogenization, cold rolling, recrystallization, and quenching. EDS and XRD checked realized chemistry and phase constitution; SEM and EBSD documented the microstructure; and quasistatic tensile and Vickers indentation measurements supplied YS and HV. Figure 2 of the main paper shows representative chemistry and microstructure verification. The main paper shows the non-uniform composition and processing coverage that motivates grouped validation.

S1.2 Audited record structure and distributions

The source workbook contains 94 records and 19 original columns. Every composition row sums to 100 at.% (Table S1). YS is available for 93 records, and HV is available for all 94. BBA09 is the only record without YS. The B batches contain 35 hardness records and 34 YS records. The C batches contain 59 records for both targets. Processing is fixed within the B campaign, whereas the C campaign varies cold work, recrystallization temperature, and hold time together with composition.

Table S1: Audited dataset statistics. Means and standard deviations are sample statistics.

Quantity	n	Min	Max	Mean $\pm$ SD
YS (MPa)	93	151.5	544.5	268.5 ± 81.6
HV	94	57.6	227.5	133.9 ± 33.6
d (μm)	94	14.66	211.75	57.39 ± 42.67
SD_{grain} (μm)	94	8.32	321.06	44.72 ± 44.83

The full dataset contains 82 unique compositions. The 93-record YS subset contains 81. Nine composition groups are repeated and account for 21 records. One composition, $\text{Co}_4\text{Cr}_4\text{Fe}_4\text{Mn}_4\text{Ni}_{72}\text{V}_{12}$, appears in both CBB and CBC, with three CBB records and one CBC record. Record-level LOO can therefore contain exact-composition overlap. LOBO reduces this overlap but does not remove it completely. We report these facts as part of the validation audit rather than treating them as another validation protocol.

Mean grain size and its standard deviation are strongly correlated ($r = 0.801$). The dimensionless coefficient of variation, $\text{SD}_{\text{grain}}/d$, ranges from 0.47 to 2.99. A claim that distribution width has an independent physical role should therefore compare raw SD with normalized width and with distribution moments such as $E[d^{-1/2}]$.

The source data also contain measurement scatter, which is distinct from grain-distribution width. SD_{YS} is available for 80 records and ranges from 0.2 to 98.7 MPa; SD_{HV} is available for all 94 and ranges from 1.43 to 12.44 HV. These values are not used as model features because replicate counts and reporting definitions are not consistent across records. In contrast, SD_{grain} is an EBSD microstructure descriptor and is evaluated explicitly as a predictor.

The strongest composition–grain-size correlations include $r(d, c_{\text{Cr}}) = +0.38$ and $r(d, c_{\text{Al}}) = -0.32$. On average, Cr-rich alloys are therefore coarser and Al-rich alloys are finer. These signs were reversed in the earlier draft. A regression of $d^{-1/2}$ on composition explains approximately 40% of its variance. Adding processing raises this value to approximately 54%. Composition and grain size are therefore not experimentally orthogonal.

Figure S1 shows the global property distributions. The gap in the grain-size histogram comes from the C-campaign processing grid and the B campaign’s fixed recipe. It is not evidence of a bimodal grain-growth mechanism. Figure S2 shows the C-campaign true stress–strain curves used to extract 0.2% offset YS. Figure S3 summarizes correlations among composition, processing, and descriptors. Several deterministic descriptors are strongly correlated with one another, so coefficient and feature-importance interpretations must account for redundancy.

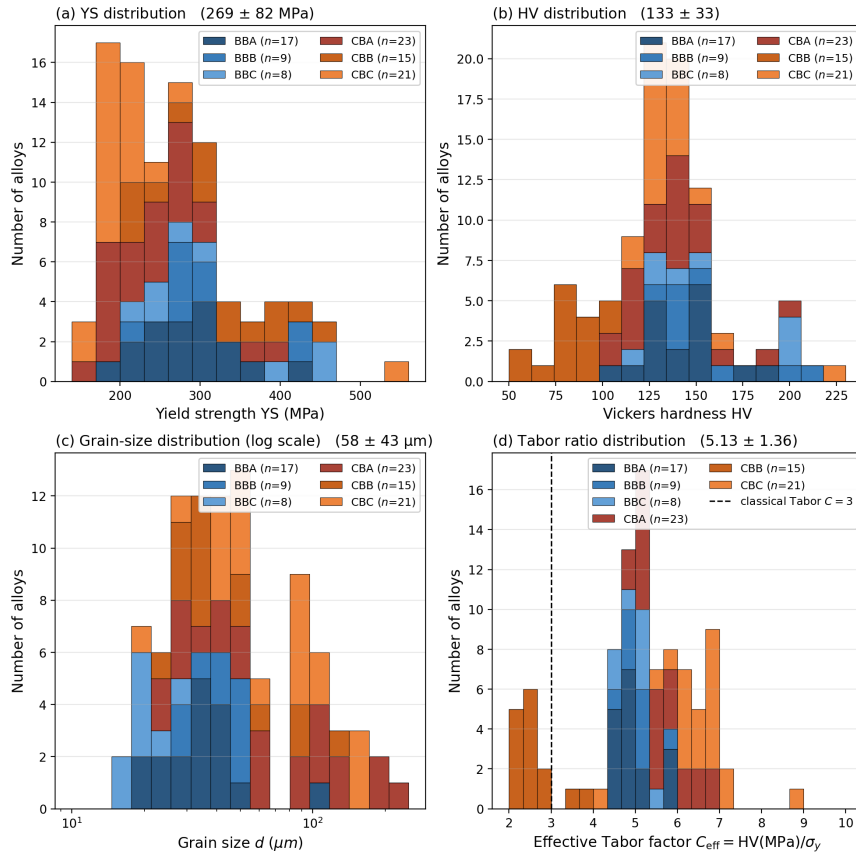


Figure S1: Audited YS, HV, grain-size, and effective hardness-to-yield distributions by batch.

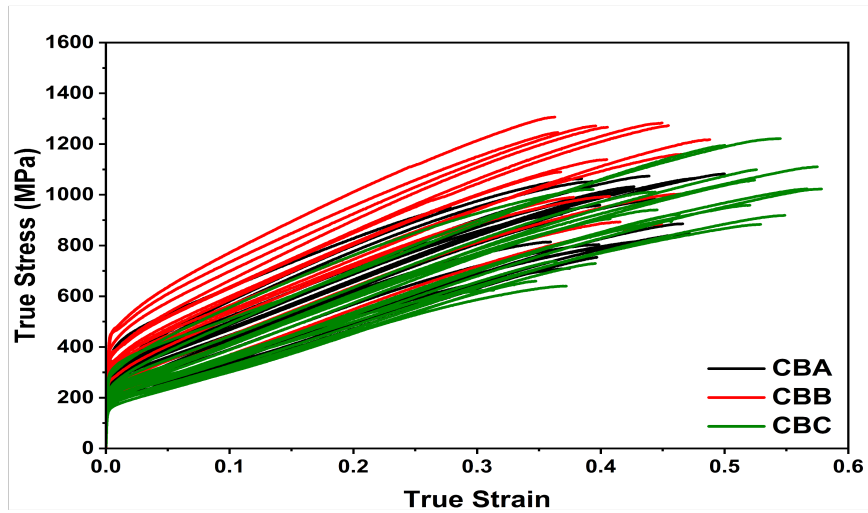


Figure S2: True stress-strain curves for the C-campaign records at a nominal strain rate of $5 \times 10^{-4} \text{ s}^{-1}$. The 0.2% offset values are the YS targets used in the analysis.

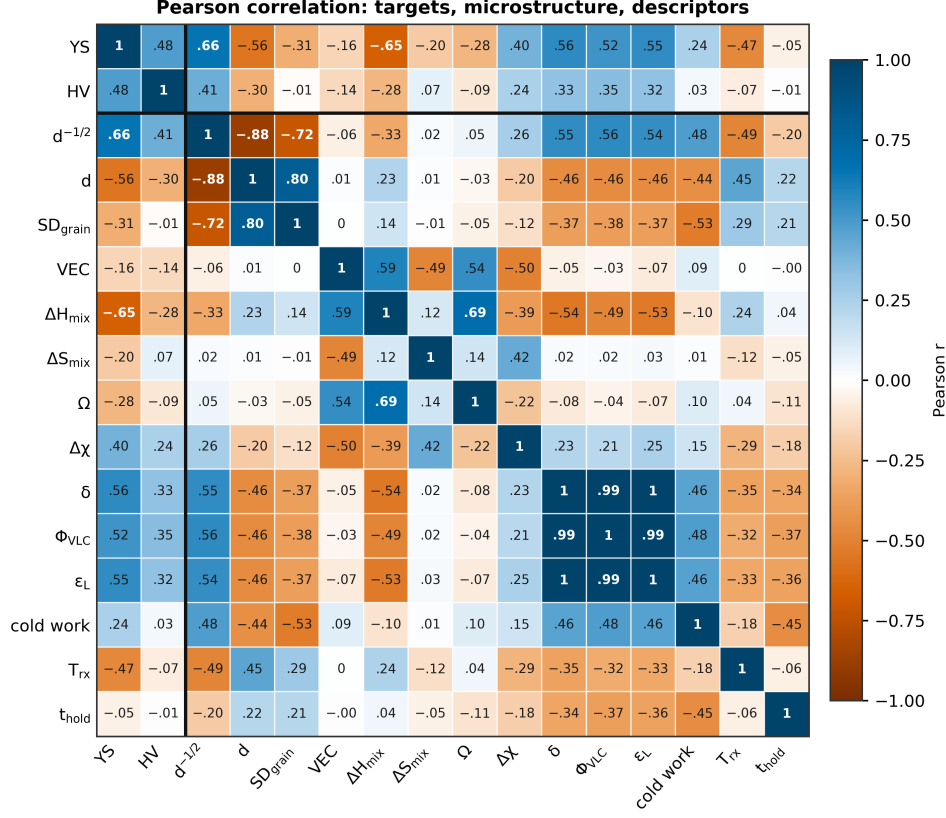


Figure S3: Pearson correlation matrix for composition, processing, grain-size, and descriptor variables. Correlation is descriptive and does not establish independent mechanistic effects.

S1.3 Grouped-validation conventions and primary results

For every protocol, predictions from all held-out records are pooled before calculating

$$Q^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}. \quad (\text{S1})$$

The manuscript reports this pooled quantity as cross-validated R^2 . It is not the mean of the fold-specific R^2 values; Table S2 lists the pooled scores for every model. Preprocessing is fit within each training fold. To calculate batch-clustered intervals, the six batches are resampled with replacement 1000 times while each batch remains intact. Because there are only six clusters, these intervals show sensitivity to the sampled batches but are not precise inferential confidence intervals.

Table S2: Primary grouped-validation results reproduced by the revision-specific validation script. The Family 5 HV result is conditional on an equation structure selected using the complete dataset before the coefficients were refit in each fold.

Target	Family/model	Status	5-fold	LOO	LOBO
YS	F1 Classical HP	baseline	0.405	0.406	0.373
YS	F2 PCA-OLS	fold-contained	0.462	0.480	0.362
YS	F3 M3	pre-specified	0.666	0.652	0.625
YS	F3 M15 SD interaction	pre-specified	0.731	0.694	0.694
YS	F4 Lasso S2	fixed settings	0.683	0.674	0.621
YS	F4 LightGBM S2	fixed settings	0.632	0.574	0.615
HV	F1 Classical HP	baseline	0.086	0.136	-0.077
HV	F4 LightGBM S1	fixed settings	0.195	0.323	0.112
HV	F5 fixed form	post-selected	0.725	0.727	0.636

The validation hierarchy changes the scientific interpretation. LOO measures record-level interpolation and can benefit from repeated or nearby compositions. LOBO tests a new iteration while retaining the practices shared across the broader program, and is therefore the primary internal grouped-transfer test. The literature stress test asks a different question about numerical behavior on heterogeneous published records, while singularity auditing checks whether a closed form remains finite and bounded over the audited domain.

S1.4 Why no hard predictive ceiling is reported

The source analysis contained two incompatible ceiling calculations. If every singleton composition is assigned zero residual, an in-sample composition-mean oracle gives approximately $R^2 = 0.86$. If variance is pooled only from repeated compositions, the within-composition variance is approximately 6979 MPa^2 . This is slightly larger than the total YS variance of 6651 MPa^2 , giving a formal value near -0.05 . Neither value is a defensible out-of-sample ceiling. There are 72 singleton YS compositions, and the repeated groups also differ in grain size and processing.

Similarly, $1 - \langle \text{SD}_{YS}^2 \rangle / \text{Var}(YS) \approx 0.97$ is not a universal noise ceiling. This calculation assumes that the reported within-alloy standard deviations are irreducible errors in the modeled target. If the target is a replicate mean, its sampling uncertainty depends on the number of replicates through $\text{SD}/\sqrt{n}$. A measurement-limited ceiling cannot be estimated without replicate-level measurements and replicate counts.

S1.5 Batch coverage in composition space

Six principal components capture 96.9% of the composition variance. In the PC1–PC2 projection, the mean pairwise convex-hull overlap is approximately 0.21 (Fig. S4). This two-dimensional diagnostic does not prove support in the full composition space, but it explains why the LOBO folds differ in difficulty. CBC is relatively isolated, while CBA is well covered by the remaining batches. The aggregate LOBO score therefore combines near-interpolation with genuine extrapolation. It should be read together with the per-batch results in Table S14.

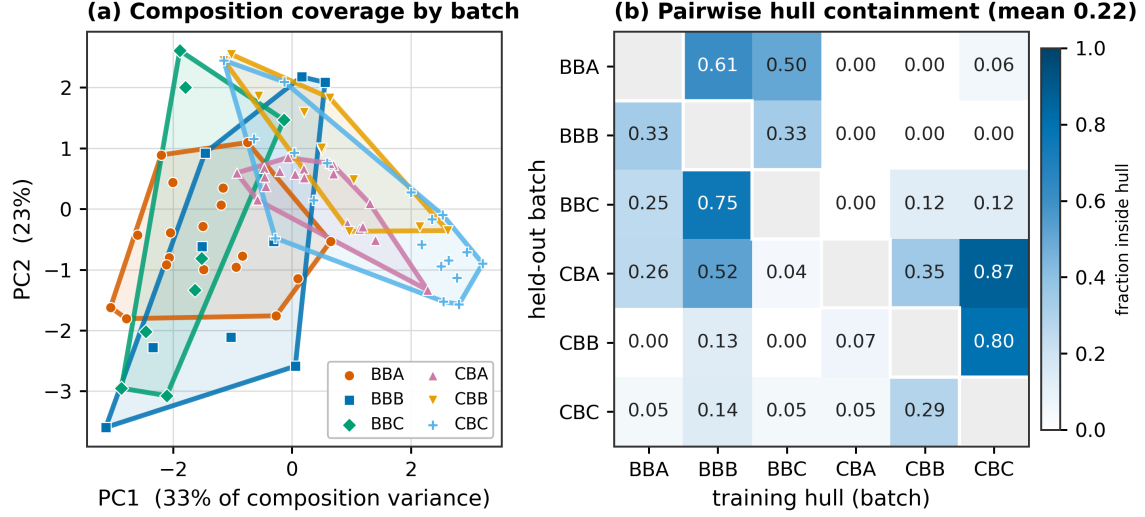


Figure S4: Batch coverage in the PC1–PC2 projection of composition space: batch convex hulls (left) and pairwise hull containment (right). These values depend on the projection and do not guarantee support in the full composition space.

S2 Family 1: Classical Hall-Petch

Nine pre-specified scaling laws were fit to YS (Table S3). Five monotonic laws fall within $\Delta\text{BIC} < 2$ of the classical Hall–Petch form. The optimized exponent is $n = 0.548$. Its Bayesian posterior is 0.53 ± 0.16 , with a 94% HDI of approximately $[0.22, 0.82]$. These results support $d^{-1/2}$ as a simple baseline but do not statistically rule out nearby exponents.

Table S3: Family 1 comparison of pre-specified grain-size laws for YS, ranked by BIC. Here, p is the number of fitted regression coefficients.

Scaling law $f(d)$	p	Train R^2	LOO R^2	LOO RMSE	BIC	ΔBIC
$d^{-1/2}$ (Hall–Petch)	2	0.429	0.406	62.5	774.5	0.0
$\ln(d)/d$	2	0.429	0.406	62.5	774.5	0.0
$d^{-2/3}$	2	0.429	0.405	62.6	774.6	0.1
$d^{-1/3}$	2	0.427	0.403	62.6	774.9	0.4
d^{-1}	2	0.419	0.395	63.1	776.1	1.6
$\ln(d)$	2	0.413	0.388	63.4	777.2	2.7
$d^{-0.548}$	3	0.430	0.406	62.5	779.0	4.5
$d^{-1/2} + d^{-1}$	3	0.430	0.395	63.1	779.0	4.5
$d^{-1} + d^{-2}$	3	0.429	0.387	63.5	779.1	4.6

Table S4: Bayesian PSIS-LOO comparison for the Family 1 YS laws. The stacking solution is reported but is not treated as proof of a unique law.

Model	elpd_{loo}	ΔLOO	p_{loo}	Weight
Hall-Petch $d^{-1/2}$	-518.5	0.0	4.2	1.000
Composite $d^{-1/2} + d^{-1}$	-518.5	0.1	4.2	0.000
Critical-thickness $\ln(d)/d$	-518.5	0.1	4.3	0.000
Intermediate $d^{-2/3}$	-518.7	0.2	4.4	0.000
Baldwin $d^{-1/3}$	-518.7	0.2	4.2	0.000
Optimized exponent d^{-n}	-518.8	0.3	4.5	0.000
Logarithmic $\ln(d)$	-519.7	1.2	3.9	0.000
Dunstan-Bushby d^{-1}	-520.3	1.8	4.3	0.000

The classical YS baseline gives LOO/LOBO $R^2 = 0.406/0.373$ (Fig. S5). For HV, the corresponding values are $0.136/ - 0.077$. Mean grain size is therefore a useful YS baseline within the sampled program, but it does not transfer reliably across batches for HV. Table S6 repeats the scaling-law comparison for hardness, and Fig. S6 shows the corresponding fits.

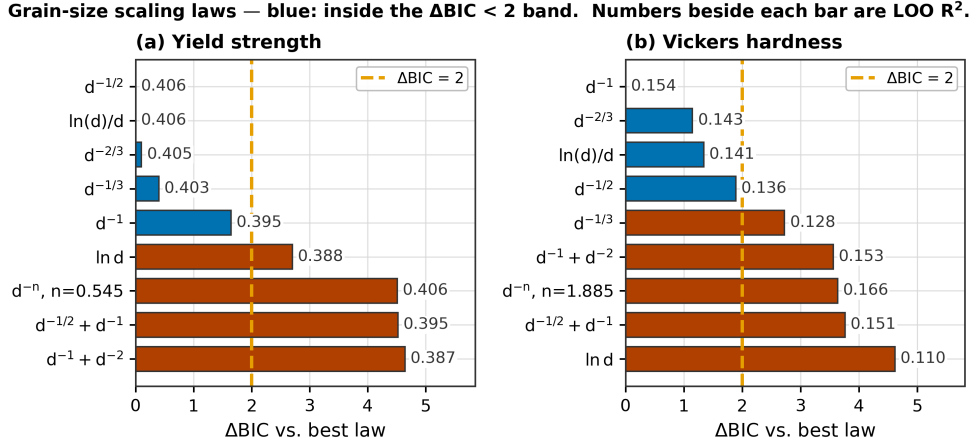


Figure S5: Information-criterion differences among pre-specified YS grain-size laws.

The Bayesian comparison is summarized in Table S4, with the YS fits shown in the main paper, the free-exponent posterior in Fig. S7, and the model weights in Fig. S8. The Bayesian stacking weight concentrates on the classical form, even though several candidate laws have almost identical expected log predictive density. We therefore do not treat this boundary solution as decisive evidence for one exponent. Parameter counting also differs across the stored tables. Some Bayesian tables include the noise parameter, while the frequentist BIC tables count regression coefficients. The revised manuscript compares models only within internally consistent tables.

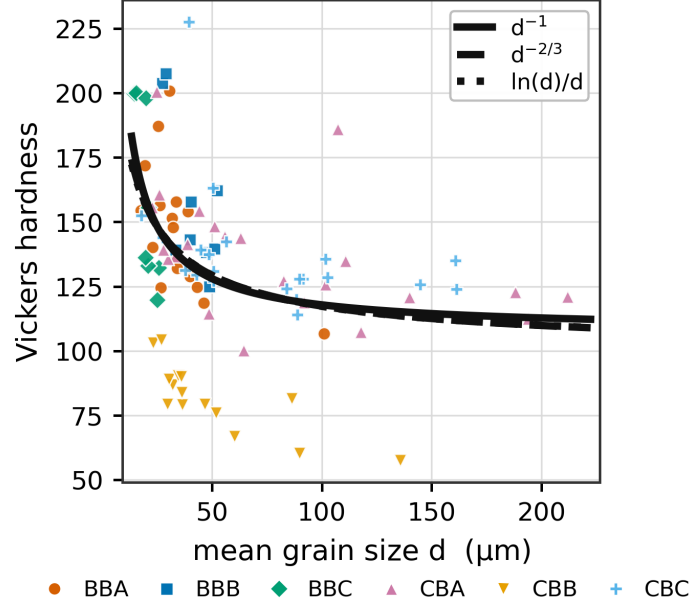


Figure S6: The three best-supported Family 1 grain-size laws fitted to HV over the observed range, with points coloured by batch. The curves overlap as closely as they do for YS, but the CBB batch lies systematically below all of them, consistent with the negative batch-level R^2 in Table S2. The corresponding YS comparison is shown in the main paper.

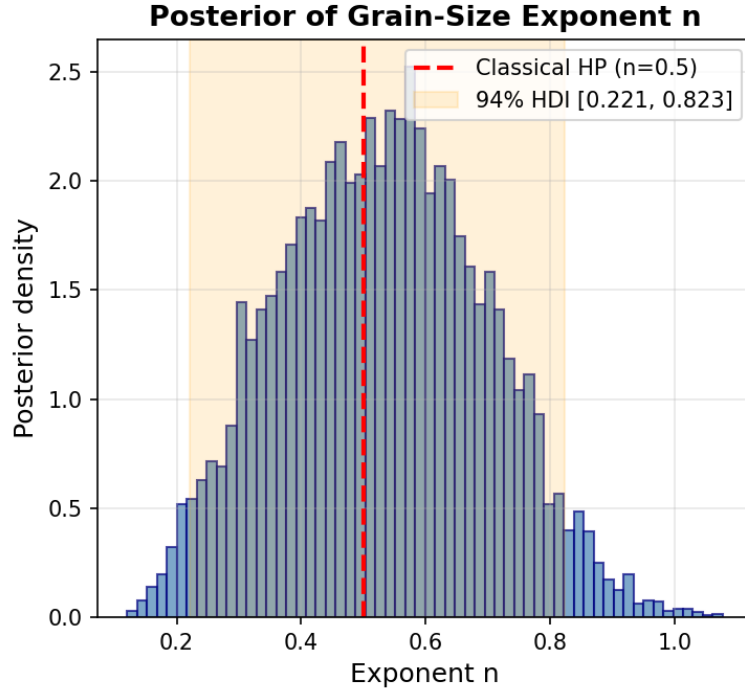


Figure S7: Posterior distribution of the free grain-size exponent n in $\sigma_y = \sigma_0 + kd^{-n}$; the dashed line marks $n = 0.5$.

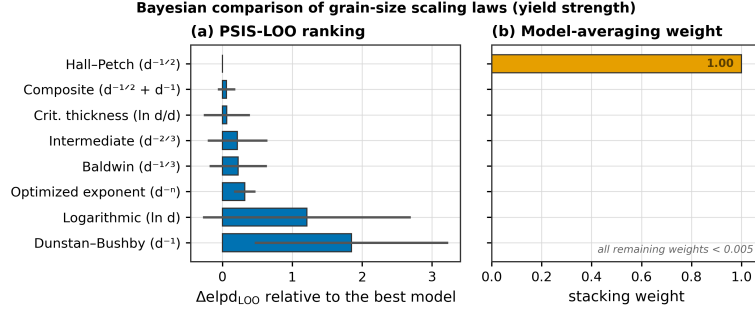


Figure S8: Bayesian model-averaged grain-only prediction with its 94% credible band. The large residual scatter shows that grain size alone does not capture the dominant property variation.

The within-replicate analysis (Table S5) gives local slopes from approximately -8539 to 9282 in Hall-Petch units. The fixed-effect pooled slope is near 2144 , with Cochran $Q \approx 251$ and $I^2 = 96.8\%$. This apparent heterogeneity cannot be attributed to composition because most groups contain only two records and processing is not held fixed. In addition, the two-stage k_{eff} regression subtracts an intercept from a model that already assumes constant k . The supported conclusion is that constant-slope M3 is the best predictive model in Family 3. This design cannot identify variation in k_{HP} .

Table S5: Descriptive within-composition fits for the nine repeated-composition groups (21 records). The values compare transformations after group demeaning but do not identify composition-specific slopes.

Scaling law	Pooled within-group R^2	R^2 for the $n = 3$ group
$d^{-1/2}$	0.778	0.995
$d^{-1/3}$	0.781	0.997
$d^{-2/3}$	0.775	0.993
d^{-1}	0.769	0.987
$\ln(d)/d$	0.776	0.993
$\ln d$	0.786	0.999

Table S6: Family 1 grain-size laws fit separately to HV. The optimized exponent was selected using the full dataset and is descriptive.

Scaling law $f(d)$	p	Train R^2	LOO R^2	ΔBIC
$d^{-1.728}$	3	0.193	0.166	0.0
d^{-1}	2	0.185	0.154	0.9
$d^{-2/3}$	2	0.175	0.143	2.1
$\ln(d)/d$	2	0.174	0.141	2.3
$d^{-1/2}$	2	0.169	0.136	2.8
$d^{-1/3}$	2	0.161	0.128	3.7
$d^{-1/2} + d^{-1}$	3	0.192	0.151	4.7
$\ln(d)$	2	0.144	0.110	5.6

S3 Family 2: Physics-derived descriptors

S3.1 Varvenne–Leyson–Curtin

Following Varvenne et al. [4, 5], the zero-temperature critical resolved shear stress is

$$\tau_{y0} = 0.051\alpha^{-1/3}\bar{\mu}\left(\frac{1+\bar{\nu}}{1-\bar{\nu}}\right)^{4/3}f_1(w_c)\left(\frac{\sum_i c_i\Delta V_i^2}{b^6}\right)^{2/3}, \quad (\text{S2})$$

and the finite-temperature polycrystal prediction is

$$\sigma_{\text{VLC}}(T) = M\tau_{y0}\left[1 - \left(\frac{k_B T}{\Delta E_b}\right)^{2/3}\right]. \quad (\text{S3})$$

The activation barrier is

$$\Delta E_b = 0.274\alpha^{1/3}\bar{\mu}b^3\left(\frac{1+\bar{\nu}}{1-\bar{\nu}}\right)^{2/3}f_2(w_c)\left(\frac{\sum_i c_i\Delta V_i^2}{b^6}\right)^{1/3}. \quad (\text{S4})$$

Here $M = 3.06$, $\alpha = 0.123$, $f_1 = 0.35$, $f_2 = 5.70$, and $b = (4\bar{V})^{1/3}/\sqrt{2}$. We use binary-solid-solution-derived volumes for Ni–Co–Fe–Cr–Mn and tabulated or extrapolated FCC volumes for Al, Cu, and V. Rule-of-mixtures elastic constants replace measured alloy values. These choices are material approximations and do not exactly reproduce a first-principles VLC calculation.

S3.2 Weighted Labusch and Toda–Caraballo adaptations

The weighted Labusch extension is

$$\sigma_{\text{Lab}} = \bar{\mu}\varepsilon_L^{4/3}c_{\text{eff}}^{2/3}, \quad \varepsilon_L = \sqrt{\sum_i c_i(\delta_{\mu,i}^2 + \alpha_L^2\delta_{r,i}^2)}, \quad (\text{S5})$$

with $\alpha_L = 16$. This multicomponent weighting is an adaptation of Labusch theory [6]. The Toda–Caraballo implementation is

$$\sigma_{\text{TC}} = M\left(\sum_i B_i^{3/2}X_i\right)^{2/3}, \quad B_i = 3\bar{\mu}\varepsilon_i^{4/3}Z, \quad (\text{S6})$$

with damped modulus and Vegard size misfits [7]. Pair-specific binary B_i values are not available for the eight-element system, so we use a shared theoretical prefactor.

The Labusch and Toda–Caraballo constants are calibrated once to reproduce 125 MPa for CoCrFeMnNi [8]. VLC is not calibrated. Table S8 separates the calibrated errors from the 2–5× errors of the uncalibrated variants.

Table S7: Cantor-anchored SSS benchmark for source alloys with traceable references in the local bibliography. Values are model outputs in MPa. $\sigma_{0,\text{exp}}$ is the reported YS after the cited Hall–Petch contribution is subtracted.

Alloy	VLC	Labusch	Toda–Caraballo	$\sigma_{0,\text{exp}}$
NiCoCr [9]	233	233	135	218
NiCoFeCr [10]	222	160	146	100
CoCrFeMnNi [8]	243	125	125	125

Table S8: Family 2 SSS predictions compared with Family 1 Hall–Petch-corrected YS for 93 alloys. Ratio is the mean prediction divided by the mean corrected response. +HP uses Ridge with the SSS estimate and $d^{-1/2}$.

Model	Mean	Ratio	Raw R^2	Spearman ρ	+HP LOO R^2	RMSE
VLC	342	1.78	−6.92	0.53	0.396	63.1
Labusch, calibrated	206	1.07	0.14	0.55	0.469	59.1
Toda–Caraballo, calibrated	274	1.42	−3.18	0.49	0.399	62.9
HP only	—	—	—	—	0.406	62.5

Table S7 lists the benchmark; the corresponding parity plots are shown in the main paper. As standalone LOO predictors, the archived SSS scores are 0.274 for calibrated Labusch, 0.091 for Toda–Caraballo, and 0.030 for VLC. All three are below the Family 1 YS baseline. When raw composition is already included, adding one of these deterministic SSS estimates changes LOO R^2 by no more than 0.002. The reported linear partial correlations after removing the composition signal are also below $|r| = 0.10$. These results show that the estimates add little information in this dataset. They do not invalidate the underlying theories outside the present input approximations.

The present data cannot show that the error comes only from the Vegard inputs. Approximated elastic properties, calibration transfer, phase or local-order effects, and the subtraction used to define experimental friction stress can all contribute. DFT-derived misfit volumes improve VLC predictions in selected FCC systems [11]. Improved inputs are therefore a clear next test, not a proven single explanation.

S3.3 Fold-contained PCA

The earlier Family 2 PCA result fit StandardScaler and PCA on all 93 records before CV (Table S9). In the revised calculation, both transformations are fit within every training fold. The model uses 11 inputs: six Wen descriptors, $d^{-1/2}$, SD_{grain} , cold work, recrystallization temperature, and hold time.

Table S9: Family 2 PCA-OLS before and after correcting preprocessing leakage.

Implementation	5-fold R^2	LOO R^2	LOBO R^2
Global scaling/PCA	0.525	0.516	0.441
Fold-contained scaling/PCA	0.462	0.480	0.362

PCA loadings from the full dataset remain descriptive only. The back-projected coefficients depend on feature scaling and are not interpreted as causal feature importance. The global-preprocessing implementation is retained only to document leakage and is not used for the revised comparison.

S4 Family 3: Composition/processing

Family 3 contains the OLS and M-model Hall–Petch hierarchy. These models add raw composition, recorded processing, and selected microstructure terms to the Family 1 baseline. M3 uses a shared k_{HP} while allowing the intercept to depend on composition. M15 extends M3 with additive SD_{grain} and the interaction $SD_{\text{grain}}d^{-1/2}$. Their coefficients are empirical within the sampled campaigns and are not elemental SSS constants.

M3 gives LOO/LOBO $R^2 = 0.652/0.625$. The modest change from LOO to LOBO supports transfer among the six sampled iterations. Because LOO can retain a repeated composition, we do not use it to claim strict prediction of unseen compositions.

Table S10 separates the value of measuring grain-distribution width from the value of its interaction with the Hall–Petch term. Adding SD_{grain} only as an independent term improves LOO but lowers LOBO. M15 improves both and is the strongest verified low-complexity YS model when distribution width is available. The interaction coefficient is 19.16 in the fitted numerical units, with $t = 4.71$ and $p = 1.0 \times 10^{-5}$. Relative to M3, M15 lowers BIC by 20.1. These are reproducible OLS results, but they do not identify a causal grain-width mechanism because d , SD_{grain} , composition, and processing are correlated.

Table S10: Verified Family 3 grain-width controls. Values are pooled out-of-fold R^2 .

Model	Parameters	5-fold	LOO	LOBO
M3	9	0.666	0.652	0.625
M3 + additive SD_{grain}	10	0.659	0.668	0.595
M15 + SD–HP interaction	11	0.731	0.694	0.694

Table S11: Family 3 M3 coefficients relative to Ni.

Parameter	Estimate (MPa)	95% low	95% high
Intercept	229.6	152.4	306.7
Al	−201.3	−955.4	552.8
Co	−187.2	−268.1	−106.3
Cr	−307.7	−588.5	−26.9
Cu	−334.0	−651.9	−16.2
Fe	−360.4	−520.1	−200.6
Mn	82.3	−117.6	282.2
V	291.3	41.0	541.5
k_{HP}	765.8	501.5	1030.0

The fitted M3 coefficients are listed in Table S11, and literature Hall–Petch slopes for comparison in Table S12. Al has a much wider interval than the named stable coefficients because it spans only 0–4 at.%. We therefore avoid stating that every M3 coefficient changes by less than $\pm 15\%$ under grain-size perturbation. That diagnostic applies specifically to V, Co, Cr, and Fe. In 1000 grain-size perturbations, the V, Co, Cr, and Fe coefficients changed by less than approximately 15%, while k_{HP} changed by approximately 39%. A 10,000-replicate row bootstrap gave standard-error ratios of 0.99–1.54 relative to OLS. These checks support numerical stability within the sampled records. However, the row bootstrap does not reproduce campaign-level sampling and does not guarantee transfer.

Table S12: Literature context for the fitted constant Hall–Petch slope, ordered by k_{HP} . All values are room temperature and, except where noted, are fitted to tensile yield strength. Coefficients obtained by converting hardness to strength are excluded, because that conversion depends on the work-hardening response rather than on a fixed factor; this also excludes survey values for pure metals that pool indentation and tensile data in a single fit [12]. Otto et al. report k_{HP} for CoCrFeMnNi falling from 538 at 77 K to 127 at 1073 K, so comparison across studies is meaningful only at matched test temperature. [‡]Fitted to compression data. The two Al_{0.3} alloys remain single-phase FCC by selected-area diffraction; their large coefficients are attributed to coherency strain from nanometre-scale Ni–Al clusters rather than to precipitates [13].

Alloy	k_{HP} (MPa· $\mu\text{m}^{1/2}$)	d range (μm)	Source
CoCrNi	265	0.29–111	Yoshida et al. [9]
CoCrFeNi	276	0.55–32.2	Yoshida et al. [14]
CoCrFeMnNi	494	4.4–155	Otto et al. [8]
CoCrFeNi	506	25–110	Li et al. [15]
CoCrFeNi	684	23–150	Jagetia et al. [13]
This work, Family 3 M3	766	14.7–212	shared fitted slope
CoCrFeNi	855	2.5–59	Wu et al. [16]
CrFeNi	966 [‡]	10–327	Schneider and Laplanche [17]
Al _{0.3} CoCrFeNi	1014	29–199	Jagetia et al. [13]
Co _{0.95} Cr _{0.8} Fe _{0.25} Ni _{1.8} Mo _{0.475}	1100	25–110	Li et al. [15]
Al _{0.3} CoFeNi	1244	38–179	Jagetia et al. [13]

Table S13 sets the present dataset against the same literature. Chemistry is recorded for every compiled entry, so grain size and hardness are the properties that limit coverage. Of the single-phase FCC records, 27 compositions carry grain size with yield strength and six carry hardness as well.

Table S13: Published coverage of the four measured properties, from the compilation of Borg et al. [18] (1545 records, 630 formulas). Counts are restricted to records labelled single-phase FCC. Chemistry is recorded for every entry, so the limiting properties are grain size and hardness. The compilation of Gorsse et al. (370 alloys) and the Chemical Data Collections database (2023) carry hardness and yield strength but no grain-size column, and COD’HEM [19], larger still at more than 4000 compositions from more than 400 papers, records neither grain size nor hardness. Borg’s per-paper extraction is not always complete, so the last column is a lower bound on what the named study reported; the aggregate columns do not depend on per-study completeness. Generated by `01_family1_grain_size/literature_coverage.py`.

Properties reported together	Records	Compositions	Studies	Largest study
Grain size + YS	91	27	25	7
Grain size + YS + hardness	8	6	3	3
This work	93	81	1	81

The grain-only Family 1 slope is approximately 1182 MPa· $\mu\text{m}^{1/2}$. Adding the composition-dependent intercept in M3 reduces the shared slope to approximately 766 MPa· $\mu\text{m}^{1/2}$. This change is consistent with composition–grain-size confounding in the global baseline. It does not show that 766 is a universal slope for FCC MPEAs.

Table S14: Family 3 M3 per-batch LOBO performance and composition-hull containment. The aggregate LOBO value in the main text is pooled; it is not the mean of these R^2 values.

Batch	n	Hull	F1 HP R^2	F3 M3 R^2	M3 RMSE
BBA	17	0.882	0.412	0.697	32.9
BBB	9	0.556	0.138	0.306	57.7
BBC	8	0.750	0.259	0.656	51.9
CBA	23	1.000	0.295	0.423	44.9
CBB	15	0.933	0.174	0.584	49.6
CBC	21	0.476	-0.038	0.409	60.4

S5 Family 4: Non-linear estimators

Family 4 compares twenty estimators on the identical S1–S4 matrices used elsewhere in this work, under nested tuning with ARMOTE-CV. Fifteen estimators (Bayesian ridge, ridge, Lasso, ElasticNet, kernel ridge, support vector regression, decision tree, random forest, extra trees, gradient boosting, XGBoost, LightGBM, CatBoost, Gaussian process regression, and a neural network) are tuned per outer fold; five (ordinary least squares, a multilayer perceptron, PCA-OLS, a mean-predicting Dummy baseline, and a stacking ensemble) are run at fixed settings within this panel. Algorithm S1 gives the nested procedure applied to every outer fold of every protocol; Tables S15 and S16 give the full model roster and search spaces; Tables S17 and S18 give the complete 5-fold and LOO results underlying the main-text summary.

Algorithm S1 ARMOTE-CV nested-tuning procedure, applied independently within each outer fold of each protocol (5-fold, LOO, LOBO)

Require: Feature matrix X , target y , model class $\mathcal{M}$, outer splitter $\mathcal{S} \in \{\text{KFold}_5, \text{LOO}, \text{LOGO}_{\text{batch}}\}$

- 1: **for** each outer fold $(X_{\text{train}}, y_{\text{train}}, X_{\text{test}}, y_{\text{test}})$ from $\mathcal{S}$ **do**
 - 2: Split $X_{\text{train}}, y_{\text{train}}$ into 5 inner folds for hyperparameter scoring
 - 3: **for** $t = 1$ to 50 **do** ▷ Optuna TPE trial, seed 42
 - 4: Sample candidate hyperparameters θ_t
 - 5: Fit $\mathcal{M}(\theta_t)$ across the 5 inner folds; record mean MSE and mean R^2
 - 6: **end for**
 - 7: $\theta^* \leftarrow \arg \max_{\theta_t} R^2$ over the Pareto front of (MSE, R^2)
 - 8: Fit feature and target **StandardScalers** on $X_{\text{train}}, y_{\text{train}}$ only
 - 9: Refit $\mathcal{M}(\theta^*)$ on the complete, scaled $X_{\text{train}}, y_{\text{train}}$
 - 10: Predict X_{test} with the refit model and scalers; invert target scaling to recover original units
 - 11: Store the study, θ^* , fitted model, and scalers for this fold
 - 12: **end for**
 - 13: **return** per-fold scores, pooled out-of-fold predictions
-

Table S15: Family 4 tuned-estimator search spaces. All ranges are sampled log-uniformly where marked (log); otherwise uniformly.

Model	Search space
Bayesian ridge	max_iter [100,500]; $\alpha_1, \alpha_2, \lambda_1, \lambda_2$ [$10^{-7}, 10^{-5}$] (log)
Support vector regression	C [0.1, 10^4] (log); γ [$10^{-4}, 1.0$] (log); ε [$10^{-3}, 0.5$] (log)
Decision tree	max_depth [5,50]; min_samples_split [2,20]; min_samples_leaf [1,20]
Random forest	n_estimators [50,200]; max_depth [5,50]; min_samples_split [2,20]; min_samples_leaf [1,10]
XGBoost	n_estimators [50,200]; learning_rate [0.01,0.5] (log); max_depth [3,10]; subsample [0.6,1.0]; colsample_bytree [0.6,1.0]
Gaussian process regression	kernel $\in$ {RBF, Matérn $\nu=0.5$, Matérn $\nu=1.5$, Matérn $\nu=2.5$, Rational Quadratic}; α [$10^{-10}, 10^{-1}$] (log)
Neural network	hidden_layers [1,4]; units (first layer) [32,128]; activation $\in$ {relu, tanh, selu}; learning_rate [$10^{-5}, 10^{-2}$] (log); fixed at 100 epochs, batch size 8
Ridge	α [$10^{-3}, 10^3$] (log)
Lasso	α [$10^{-4}, 10$] (log)
ElasticNet	α [$10^{-4}, 10$] (log); l1_ratio [0.05,0.95]
Kernel ridge	α [$10^{-3}, 10$] (log); γ [$10^{-4}, 1.0$] (log); kernel fixed at RBF
Extra trees	n_estimators [50,400]; max_depth [5,50]; min_samples_split [2,20]; min_samples_leaf [1,10]
Gradient boosting	n_estimators [50,400]; learning_rate [0.01,0.3] (log); max_depth [2,8]; subsample [0.6,1.0]
LightGBM	n_estimators [50,700]; max_depth [2,6]; learning_rate [0.005,0.2] (log); num_leaves [7,63]; min_child_samples [5,30]; subsample [0.5,1.0]; colsample_bytree [0.3,1.0]
CatBoost	iterations [50,500]; depth [3,10]; learning_rate [0.01,0.3] (log); l2_leaf_reg [1.0,10] (log)

Table S16: Family 4 fixed-settings estimators (no Optuna search within this panel).

Model	Configuration
Ordinary least squares	No hyperparameters
Multilayer perceptron	Hidden layers (64, 32); max_iter 2000; early stopping enabled
PCA-OLS	6 principal components + OLS; skipped for S1 (2 features, below the 6-component minimum)
Dummy	Predicts the training-partition mean; baseline for all comparisons
Stacking	Base learners: random forest, XGBoost, LightGBM, each at library defaults; meta-learner: ridge with α selected by internal CV over 20 log-spaced points in [$10^{-3}, 10^3$]

The Bayesian information criterion is reported only for the five linear-parametric estimators above that admit a closed-form effective-parameter count – ordinary least squares, ridge, Lasso, ElasticNet, and PCA-OLS (k = number of retained features +1, or $6 + 1$ for PCA-OLS) – computed from the pooled leave-one-out out-of-fold error. It is left blank for every tree-based, kernel-based, boosting, or ensemble estimator.

Four estimators are omitted from the main-text figure and comparison, for two distinct reasons. Lasso’s tuned α range, applied to a standardized target, drives all coefficients to exactly zero at S2 and S3, collapsing it to an intercept-only fit; its LOBO R^2 (-0.122) is bit-for-bit identical to the Dummy baseline at both tiers. Ridge, Bayesian ridge, and ElasticNet do not show this behavior, since their L2 or blended L1/L2 penalties shrink coefficients rather than zeroing them outright. CatBoost and decision tree are omitted as redundant, given random forest, extra trees, gradient boosting, XGBoost, and LightGBM already span the tree-ensemble and boosting space. The multilayer perceptron is omitted for instability: its scores swing between competitive and severely negative across tiers and protocols (YS S1 LOBO = -1.03 ; HV S3 LOBO = -2.28), unlike the neural network estimator (NNR), which shows similar volatility but is tuned per fold and is retained.

For YS, ridge and Bayesian ridge lead at S2 (5-fold/LOO/LOBO $R^2 = 0.695/0.662/0.614$ and $0.685/0.666/0.613$); reloading the stored per-fold Bayesian-ridge models and scalers and re-predicting each held-out batch reproduces this score exactly, with pooled RMSE = 50.5 MPa. Kernel ridge is the closest non-linear competitor ($0.695/0.660/0.594$); tree-based and boosted estimators trail further behind under LOBO specifically. The same ordering holds at S1, S3, and S4 (Tables S17, S18).

For HV, kernel ridge is the only estimator among all twenty with a positive LOBO score at S1 ($0.228/0.466/0.190$); every other estimator is negative at this tier. Once composition or processing descriptors are added (S2–S4), the record-level and batch-level pictures diverge sharply – for example, at S2, LightGBM reaches LOO = 0.520 but LOBO = -0.911 , and kernel ridge reaches LOO = 0.449 but LOBO = -0.559 – and no estimator at S2, S3, or S4 beats the Dummy baseline under LOBO.

Table S17: Family 4, 5-fold CV R^2 across the S1–S4 ladder. Models and tiers match Fig. ??; PCA-OLS is not run at S1 (fewer than 6 features).

Model	YS				HV			
	S1	S2	S3	S4	S1	S2	S3	S4
Ridge	0.384	0.695	0.653	0.646	−0.085	−0.187	−0.071	−0.134
Bayesian ridge	0.409	0.685	0.652	0.665	−0.111	−0.192	−0.076	−0.222
OLS	0.412	0.688	0.649	0.561	−0.149	−0.384	−0.172	−0.364
PCA-OLS	n/a	0.632	0.438	0.588	n/a	−0.086	−0.496	−0.317
ElasticNet	0.204	0.657	0.377	0.190	0.005	−0.112	−0.109	−0.248
Kernel ridge	0.382	0.695	0.646	0.650	0.228	0.025	0.335	0.214
Stacking	0.306	0.611	0.616	0.586	−0.024	−0.347	−0.310	−0.399
NNR	−0.867	0.661	0.626	0.687	−0.315	0.111	0.364	−0.133
Extra trees	0.330	0.638	0.617	0.600	0.184	−0.327	−0.079	−0.133
LightGBM	0.314	0.514	0.512	0.552	0.061	−0.015	0.062	0.037
Random forest	0.328	0.588	0.562	0.554	0.156	−0.147	−0.151	−0.201
GPR	0.323	0.612	0.578	0.582	0.059	−0.034	0.079	−0.032
XGBoost	0.219	0.564	0.571	0.596	−0.206	−0.065	−0.180	−0.140
Gradient boosting	0.370	0.575	0.562	0.564	0.202	−0.196	−0.286	−0.279
SVR	0.184	0.325	0.364	0.610	−2.828	−0.203	−2.205	−0.100

Table S18: Family 4, LOO R^2 across the S1–S4 ladder. Models and tiers match Fig. ??; PCA-OLS is not run at S1 (fewer than 6 features).

Model	YS				HV			
	S1	S2	S3	S4	S1	S2	S3	S4
Ridge	0.451	0.662	0.601	0.599	0.242	0.147	0.258	0.231
Bayesian ridge	0.447	0.666	0.620	0.612	0.245	0.199	0.254	0.225
OLS	0.451	0.672	0.622	0.531	0.232	0.139	0.229	0.176
PCA-OLS	n/a	0.604	0.480	0.540	n/a	0.108	−0.070	−0.032
ElasticNet	0.446	0.369	0.087	0.023	0.158	0.026	0.040	0.100
Kernel ridge	0.411	0.660	0.608	0.601	0.466	0.449	0.506	0.399
Stacking	0.290	0.604	0.592	0.578	0.213	−0.147	−0.165	−0.234
NNR	−1.609	0.650	0.563	0.518	0.233	0.422	0.363	0.194
Extra trees	0.351	0.599	0.576	0.543	0.345	0.336	0.367	0.298
LightGBM	0.271	0.565	0.566	0.560	0.217	0.520	0.457	0.432
Random forest	0.274	0.563	0.547	0.554	0.309	0.276	0.269	0.280
GPR	0.311	0.617	0.534	0.545	0.239	0.337	0.286	0.199
XGBoost	0.220	0.561	0.532	0.493	0.011	0.405	0.367	0.284
Gradient boosting	0.279	0.568	0.522	0.504	0.308	0.280	0.367	0.421
SVR	0.217	0.567	0.460	0.509	−0.554	−0.072	−0.088	−0.163

S6 Family 4: Non-linear estimators

The equal-footing panel gives every estimator the same S1–S4 input matrices and fixed splits. S1 contains both $d^{-1/2}$ and SD_{grain} , unlike the Family 1 classical baseline, which contains only $d^{-1/2}$. For YS, an S1 linear model gives LOO/LOBO $R^2 = 0.451/0.394$, compared with $0.406/0.373$ for Family 1. The larger M15 gain therefore comes from the interaction, not simply from adding the SD feature. At S2, the Lasso linear control and LightGBM have similar LOBO values of 0.621 and 0.615. For HV, adding SD linearly changes LOO/LOBO from $0.136/−0.077$ to $0.232/−0.176$, while LightGBM S1 gives $0.323/0.112$. Thus the LightGBM-to-classical comparison changes both features and estimator. Descriptor-rich input sets are generally negative under LOBO. These are comparisons within a shared Family 4 panel; they do not imply that every repository analysis uses identical inputs.

The primary audit in the main paper uses declared, fixed settings with no data-driven hyperparameter selection. Lasso uses $\alpha = 1.0$. LightGBM uses 400 estimators, a maximum depth of 3, a learning rate of 0.05, and a feature fraction of 0.8. These settings were chosen for this study and are not library defaults. The batch-bootstrap LOBO intervals overlap for YS Lasso ($[0.516, 0.664]$) and LightGBM ($[0.474, 0.664]$). This overlap does not establish equivalence, but it provides no evidence that non-linearity improves the matched S2 comparison.

The nested ARMOTE-CV panel gives YS Bayesian ridge at S2 with $R^2 = 0.685$ under 5-fold CV, 0.666 under LOO, and 0.613 under LOBO. Its generator stores one Optuna study, one selected parameter set, and one pair of feature and target scalers for every fold; across the six LOBO folds the five tuned Bayesian-ridge hyperparameters take six distinct values, and the scalers are fitted on each training split alone. Reloading those objects and predicting each held-out batch reproduces the six fold scores exactly and gives a pooled LOBO Q^2 of 0.613 with $RMSE = 50.5$ MPa. The nested implementation is therefore verified rather than assumed.

S6.1 Legacy 17-model panel and SHAP audit

The legacy 17-model panel tuned selected models on the full dataset before LOO. It also gave the tree models up to 64 interaction features. Its XGBoost scores, 0.729 under LOO and 0.574 under LOBO, therefore do not enter the primary comparison. The 0.155 difference is a validation gap; it does not prove that batch patterns cause 21% of the explained variance. Table S19 is

retained for provenance. It should not be compared numerically with the fixed-setting primary panel as though both used the same selection protocol.

Table S19: Archived legacy panel ranked by LOO R^2 . INT-ALT is a 64-feature interaction matrix, and COMPACT is a 20-feature matrix selected using the full dataset. Effective parameter counts and BIC are omitted because their definitions differed across estimator classes.

Model	Features	p	LOO R^2	LOBO R^2
XGBoost	INT-ALT	64	0.729	0.574
Average ensemble	meta	5	0.700	—
Stacking (Ridge)	meta	5	0.698	0.707
Random forest	INT-ALT	64	0.675	0.623
LightGBM	INT-ALT	64	0.673	0.625
SVR (RBF)	COMPACT	20	0.654	0.497
Family 3 M3	HP-PHYS	9	0.652	0.625
XGBoost compact	COMPACT	20	0.650	0.580
Ridge	PHYS-ALT	33	0.638	0.651
ElasticNet	PHYS-ALT	33	0.637	0.648
CatBoost	INT-ALT	64	0.637	0.515

The stacking result has an additional leakage risk. Its meta-features are LOO predictions from base models that retained other records from the held-out batch. In the legacy XGBoost model, SHAP ranks Ω , $\text{Mn} \times d^{-1}$, $\text{V} \times d^{-1}$, alternative grain-size transforms, and SSS descriptors highly (Fig. S9). These attributions describe the archived fitted model. Because the features are correlated and include constructed interactions, the SHAP values cannot identify independent physical mechanisms or a composition-dependent Hall–Petch slope.

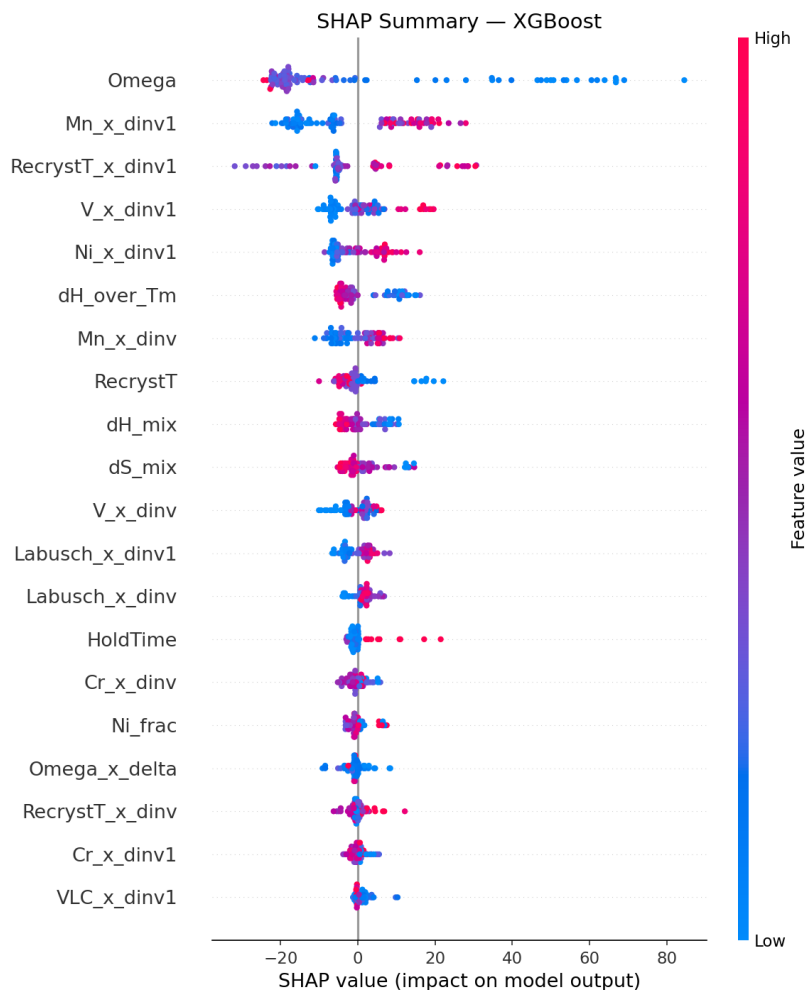


Figure S9: SHAP summary for the archived XGBoost model. The values explain this predictor, which was tuned using the full dataset; they are not causal feature effects.

S7 Family 5: Symbolic regression

S7.1 SISSO configuration and expanded controls

The archived SISSO input space contains 27 primary features. These include composition-weighted statistics of atomic radius, shear modulus, electronegativity, and melting point; thermodynamic descriptors; three SSS-related quantities; and $d^{-1/2}$ [20, 21]. Two algebraic expansion tiers using $+$, $-$, $\times$, and $\div$ are followed by sparse selection at descriptor dimension $D = 3$. The “SISSO Full” expression was selected from this space. The “SISSO Robust” control removes the small-denominator δ_μ term before the search. Here, “Robust” means numerically bounded over the audited domain, not validated transfer.

An expanded archived search added x^2 , $\sqrt{x}$, and x^{-1} , nine alternative grain-size transforms, a larger screening threshold, and BIC selection over $D = 1, \dots, 4$. Its reported LOO/LOBO $R^2 = 0.604/0.350$ was lower than the constrained-search result because the selected transforms changed across folds. An EML universal-operator control [22] gave LOO $R^2 = 0.321$ at depth one and became unstable at greater depth. A recalibrated pure-metal descriptor relation [23] gave LOO $R^2 = 0.346$. These controls support the hypothesis that a useful inductive bias can help. However, the archived records do not show that every structure-selection decision was nested within each outer split.

S7.2 PySR search grid

PySR used three feature pools, renamed P1–P3 here to avoid confusion with the five Family labels. P1 contains $d^{-1/2}$ and SD_{grain} . P2 adds raw composition and processing. P3 instead adds six Wen descriptors and processing. The archived files call these pools F1–F2. The operator sets were O1 (+, −), O2 (+, −, ×, ÷), and O3 (O2 plus protected square root, logarithm, square, and cube). Each cell used 50–75 search iterations. Equation structures and elbow or accuracy selections were obtained using the complete target dataset.

The full PySR grid is given in Table S20; its complexity–loss front is shown in the main paper. For CV, the numerical constants in the selected full-data expression were replaced with free parameters and refit in each training fold. This tests a frozen structure after it has been selected; it does not repeat the structure search. The stored LOBO values in Table S21 are averages of per-batch R^2 . In contrast, the values reported in the main text are pooled Q^2 . We report the stored values for provenance but do not place them on the same leaderboard.

Table S20: Archived PySR grid selected at the elbow. Stored LOBO uses the archived fold aggregation. Ext. is RMSE on the heterogeneous E1–E3 literature stress set, and Safe records the archived singularity audit. P1–P3 are feature pools within Family 5.

Target	Cell	Const.	LOO R^2	Stored LOBO	BIC	Ext. RMSE	Safe
YS	P1–O1	1	0.019	−0.251	818	112	yes
YS	P1–O2	2	0.482	0.276	761	168	yes
YS	P1–O3	1	0.399	0.205	773	158	yes
YS	P2–O1	1	0.035	−0.281	817	108	yes
YS	P2–O2	2	0.605	0.475	737	114	yes
YS	P2–O3	2	0.604	0.496	737	108	yes
YS	P3–O1	1	0.290	0.068	788	117	yes
YS	P3–O2	2	0.632	0.500	727	133	yes
YS	P3–O3	2	0.674	0.555	716	128	yes
HV	P1–O1	1	−0.018	−4.16	664	32	yes
HV	P1–O2	1	−0.020	−4.16	664	33	yes
HV	P1–O3	3	0.579	−1.15	587	9944	no
HV	P2–O1	1	−0.014	−4.15	663	32	yes
HV	P2–O2	3	0.568	−1.23	587	97	yes
HV	P2–O3	4	0.678	−0.48	562	17895	no
HV	P3–O1	1	0.054	−4.62	657	37	yes
HV	P3–O2	3	0.631	−0.87	570	75	yes
HV	P3–O3	4	0.681	−2.79	561	51	yes

Table S21: Selected PySR YS equations. CV values are post-selection fixed-form scores. Stored LOBO is the mean of the fold-wise R^2 values. Every external R^2 is negative.

Cell	Selection	LOO R^2	Stored LOBO	Ext. RMSE	Ext. R^2
P2–O2	accuracy	0.808	0.696	129.4	−0.496
P2–O3	accuracy	0.808	0.689	111.9	−0.119
P3–O3	accuracy	0.765	0.668	120.4	−0.295
P3–O3	elbow	0.674	0.555	128.4	−0.474

PySR BIC counts the fitted numerical constants but not the many variables, operators, and structures that were attempted. We therefore do not compare it with the BIC of a pre-specified linear model set. During external evaluation, the protected functions are $\sqrt{|x|}$ and $\log(|x| + 10^{-12})$. The printed equation must include these protections; otherwise the displayed and evaluated forms are different. Protection keeps a square root real but does not remove a pole when that root is in a denominator. The P3–O3 accuracy equation contains

$\text{SD}_{\text{grain}}/\sqrt{|\Delta S_{\text{mix}} - 8.2739|}$. The internal minimum $\Delta S_{\text{mix}} = 8.3637$ is only 0.0898 above the singular value, so this expression must not be extrapolated without a stated descriptor-domain check.

SISSO Full has the form

$$\sigma_y = 120.5 \frac{\sigma_r^2}{r_{\text{range}}} + 9356 \frac{d^{-1/2}}{\Delta S_{\text{mix}}} + 1134 \frac{\sigma_\chi^2}{\delta_\mu} - 43.3. \quad (\text{S7})$$

The δ_μ denominator becomes small when the constituents have similar shear moduli. The aggregate external RMSE is 420.6 MPa. The bounded SISSO Robust variant removes this ratio and gives 162.9 MPa. This 61% reduction in RMSE does not establish deployability because $R^2 = -1.37$.

S7.3 Corrected HV fixed form

The refit hardness form is summarized in Table S22. The previously printed equation had two coefficient signs opposite to those in the fitted model. When evaluated as printed, it gives full-data $R^2 = -29.81$ and RMSE 185.4 HV. The corrected full-data fit is

$$H_V = 222.306 + 84.479 \frac{6.93 - d}{\text{SD}_{\text{grain}}} - 1.0046 \frac{\Delta H_{\text{mix}}}{t_{\text{hold}}^2}. \quad (\text{S8})$$

Table S22: Performance of the corrected HV fixed form when its coefficients are refit within every fold.

Protocol	R^2	RMSE (HV)
Full fit	0.744	16.90
5-fold	0.725	17.50
LOO	0.727	17.46
LOBO	0.636	20.16

A 10,000-resample bootstrap gives $c_0 = 221.82 \pm 9.64$, $c_1 = 84.12 \pm 7.34$, and $c_2 = -1.006 \pm 0.106$ as the bootstrap mean $\pm$ SD. Here c_0 and c_1 have units of HV, and c_2 has units of $\text{HV} \cdot \text{h}^2 \cdot \text{mol kJ}^{-1}$. The 95% interval excludes zero for each coefficient. These intervals are conditional on the selected equation structure and therefore do not include structure-selection uncertainty.

The fixed form predicts held-out batches within the combined program, but its structure was selected after the full-data search fronts were examined. Hold time is 0.5 h for every B-campaign record and is 1, 2, 4, or 8 h in Campaign C. It separates campaign with $\text{AUC} = 1.000$, and $\Delta H_{\text{mix}}/t_{\text{hold}}^2$ gives $\text{AUC} = 0.970$. No hold-time effect is identifiable within Campaign B. By comparison, the grain-width term $(6.93 - d)/\text{SD}_{\text{grain}}$ gives $\text{AUC} = 0.533$ and is nearly campaign-neutral. The hold-time term is therefore a campaign proxy in this dataset, whereas the grain-width term is a separate post-selected microstructure hypothesis. The equation is a prospective hypothesis, not yet a deployable hardness law.

The zero at $d = 6.93 \mu\text{m}$ is not a singularity. Singularities occur when $\text{SD}_{\text{grain}} = 0$ or $t_{\text{hold}} = 0$. In the observed data, every d exceeds $14.66 \mu\text{m}$, $\text{SD}_{\text{grain}} \geq 8.32 \mu\text{m}$, and $t \geq 0.5 \text{ h}$. The audit script evaluates four registered equations over the observed and external values. It does not sweep every closed form over a multivariate convex hull.

S8 Literature stress test by evidence class

The literature table combines records with different target provenance. Class E1 contains 54 direct strength values: 48 records from the Borg/Citrine compilation [18] and six compression

records for CrFeNi from Schneider and Laplanche [17]. These records represent only 18 exact compositions and mix sources and test modes. The legacy extraction restricted d to 3–500 μm , removed YS values above 800 MPa, and omitted one large-grain CrFeNi record because it departed from the expected Hall–Petch trend. Because these rules use outcome or relationship information, E1 is a heterogeneous stress test rather than an unbiased external benchmark.

Class E2 contains three YS values reconstructed from published Hall–Petch fits [8]. Class E3 contains 25 YS proxies converted from HV [24] using C_{eff} inferred from the present dataset. E2 and E3 are therefore model-derived evidence, not independent measurements of the target. With only three values, the E2 R^2 entries in Table S23 are descriptive and are not interpretable as generalization estimates. These evidence classes do not change the five model families, and they are never pooled to inflate the apparent number of direct external measurements.

Table S23: Literature stress-test metrics separated by evidence class. Negative R^2 on E1 means the model does not beat the E1 mean baseline.

Class	Model family and form	n	R^2	RMSE (MPa)
E1 measured YS	F3 M3	54	−0.508	149.5
E1 measured YS	F5 SISSO Robust	54	−0.726	160.0
E1 measured YS	F5 SISSO Full	54	−5.084	300.3
E2 HP-derived	F3 M3	3	0.677	49.0
E2 HP-derived	F5 SISSO Robust	3	0.849	33.5
E3 HV-converted	F3 M3	25	−2.157	96.9
E3 HV-converted	F5 SISSO Robust	25	−9.642	177.9
E1–E3	F3 M3	82	−0.580	132.9
E1–E3	F5 SISSO Robust	82	−1.373	162.9
E1–E3	F5 SISSO Full	82	−14.813	420.6

The direct subset is the primary literature result, and no model beats its mean baseline. Aggregate RMSE values can compare numerical robustness on the same heterogeneous compilation. They cannot establish deployability or unbiased external accuracy. A future external benchmark should be defined before outcomes are inspected and should use consistent tensile definitions. It should also include complete metadata for composition, processing, phase, grain statistics, temperature, and test mode.

S9 Hardness–yield correspondence

Tabor’s empirical Vickers relation is written as $H \simeq 3\sigma_f(\varepsilon_r)$ with $\varepsilon_r \approx 0.08$ [25]. Instrumented-indentation formulations, including that of Dao et al. [26], use different definitions of representative strain. The value 0.08 is therefore not universal. With $C_{\text{eff}} = H/\sigma_y$ and a Hollomon assumption between 0.2% and 8% strain,

$$n_{\text{app}} = \frac{\ln(C_{\text{eff}}/3)}{\ln(0.08/0.002)} = \frac{\ln(C_{\text{eff}}/3)}{\ln 40}. \quad (\text{S9})$$

The dataset gives $C_{\text{eff}} = 5.13 \pm 1.36$ and $n_{\text{app}} \approx 0.15 \pm 0.07$, where the latter uncertainty is propagated from the reported C_{eff} summary. Offset yield is not a measured flow stress at exactly 0.002 true plastic strain, and the calculation assumes Hollomon behavior. The result is therefore an apparent index rather than a measured tensile exponent.

Global Spearman $\rho(\text{HV}, \text{YS}) = 0.46$, while the per-batch values range from 0.70 to 0.95. The association becomes weaker but does not reverse. The revised paper therefore uses “aggregation-induced attenuation” rather than “Simpson’s paradox.” The partial rank correlation after controlling for d is 0.24. This single adjustment does not identify V, or any other composition

variable, as a causal scrambler. We do not recommend the fitted V correction without prospective validation.

The archived descriptive regression is

$$\log C_{\text{eff}} = 1.36 + 0.10 \log d - 2.06x_V, \quad (\text{S10})$$

where d is entered in the numerical units used for fitting. This regression describes an association in the present campaigns. It is neither dimensionally invariant nor mechanistically identified. Agreement at the top of the distribution also depends on the threshold. Two of the five hardest records are among the five strongest; the overlap increases to 8/10 and 11/20 for broader thresholds. These results support using HV for ranking only within a narrow composition window with similar processing.

S10 Reproducibility across model families

The scripts `06_validation/verified_analysis.py`, `06_validation/grouped_validation.py`, and `03_family3_composition_processing/sdgrain_models.py` in `scripts/` generate the revision-specific audit files. The second script writes `results/main2_grouped_validation.csv`, which contains the three internal protocols used in the main paper: 5-fold, LOO, and LOBO. The third writes `results/sdgrain_m15_validation.csv`, which independently refits M3, the additive SD control, and M15. Other verified files contain the fold-contained Family 2 PCA predictions, corrected Family 5 hardness coefficients and bootstrap intervals, the cross-batch composition audit, the E1–E3 external evidence classes, the singularity audit, and a method-claim audit.

Family 3 M15 OLS coefficients, information criteria, and grouped-CV scores are reproducible from the prepared dataset. Its earlier Bayesian stacking weight and PSIS-LOO comparison remain archival because that probabilistic analysis was not regenerated. The Family 4 nested ARMOTE-CV scores are reproducible from the stored per-fold studies, parameters, models, and scalars. Family 5 PySR remains exploratory until equation discovery is repeated within every outer training split. Full-data loadings, feature rankings, and symbolic search fronts are descriptive and should not be substituted for held-out performance.

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